Determinants of hyperparameter robustness in connectome reservoir computing

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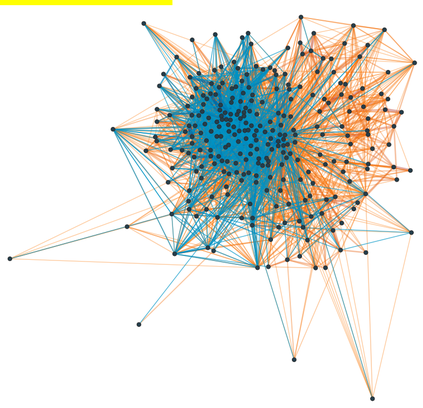
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MAIN FINDING

Higher task-agnostic performance accompanies greater sensitivity to hyperparameter choice.

The empirical *C. elegans* reservoir occupies a relatively low-variance regime.

01 Controlled reservoirs



***C. elegans*: 297 neurons, 3,604 chemical edges.** Elegan-Sign [1].
Orange: positive Blue: negative
Largest connected component shown; arrows omitted.

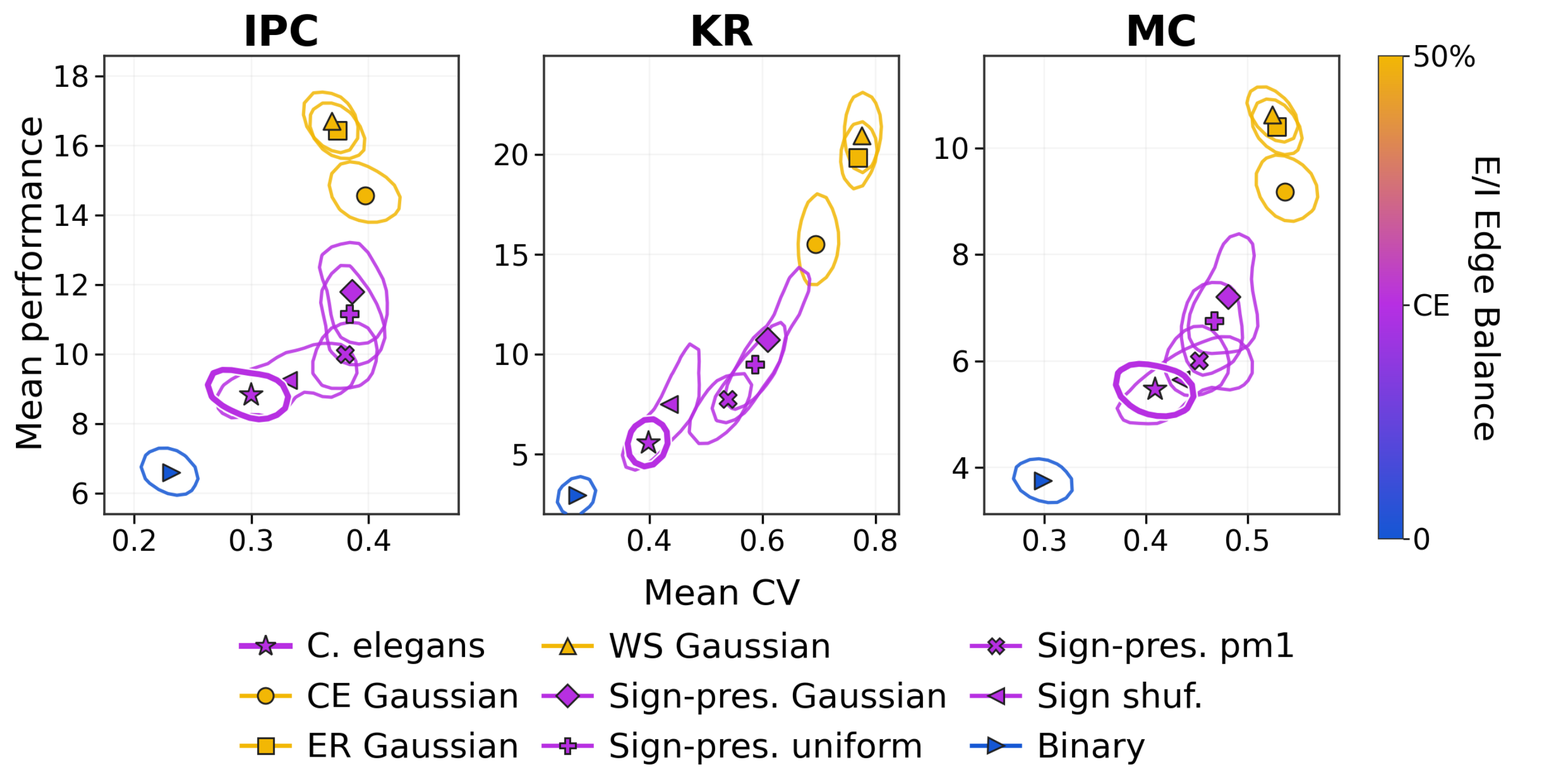
Fixed recurrent weights; only the readout is trained. Perturb topology, signs, and weights, then normalize spectral radius.

96 settings × 200 trials
Vary target radius, leak rate, input scale, and bias.

MC: recall of the past 30 inputs.
IPC: nonlinear recall, single-delay polynomial orders 1–5 [2].
KR: effective dimensionality of responses to distinct histories [3].

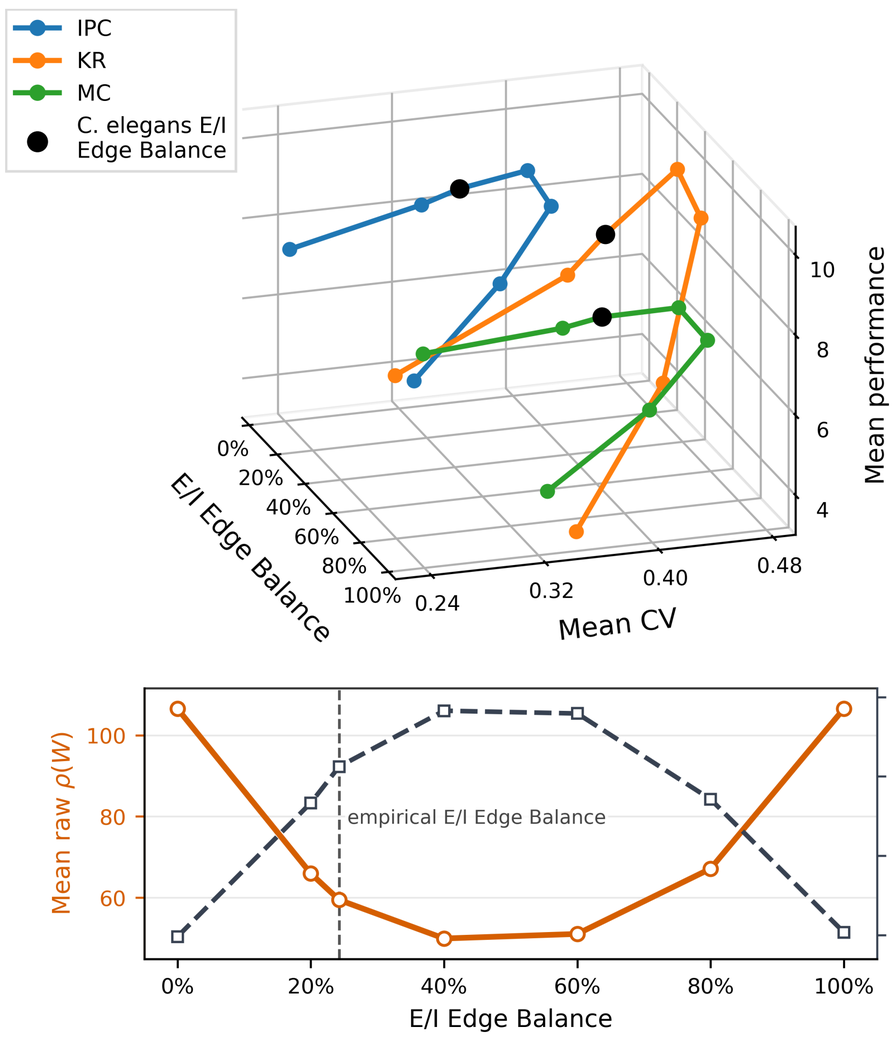
$\mathrm{CV} = \frac{\mathrm{sd}}{\mathrm{mean}}$ across the 96 settings.
Lower CV = greater robustness.

02 Performance comes with sensitivity



Higher and farther right: more performance, less robustness. Contours enclose the highest-density 50% of trials; markers identify architectures. Color indicates the negative-edge fraction: 0%, empirical $\approx 24.25\%$, or 50%.
CE: *C. elegans*; ER: Erdős–Rényi; WS: Watts–Strogatz; pm1: signed-unit weights. E/I balance refers to **edges**, not neuron counts.

03 E/I balance traces the tradeoff



Changing signs moves performance and CV together. Top: IPC, KR, and MC across the negative-edge fraction; black markers denote the empirical fraction. Bottom: raw radius and CV averaged across metrics. Points average 200 trials.

04 What the curves show

Connectivity and weight magnitudes stay fixed while signs are randomly reassigned at each negative-edge fraction.

Near the all-positive and all-negative endpoints: lower performance and lower CV, indicating greater hyperparameter robustness.

Toward 50% negative edges: higher performance and higher CV, indicating greater sensitivity to hyperparameter choice.

Why normalization matters

Every perturbed matrix W' is rescaled:

$$W^{\mathrm{res}} = \frac{\rho_{\mathrm{target}}}{\rho_{\mathrm{raw}}} W', \qquad \rho_{\mathrm{raw}} = \max_i |\lambda_i(W')|.$$

The raw radius decreases toward the balanced regime. Reaching the same target radius therefore requires a **larger scaling factor** in this regime.

These curves combine changes in sign balance and placement with **normalization-induced rescaling**. They do not isolate a causal effect of E/I balance alone.

Report raw spectral radius alongside architecture comparisons.

The full paper finds the same qualitative tradeoff in an edge-matched random topology and after removing unknown or complex-sign edges.

Scope and limitations. Robustness here means low relative variation across the tested hyperparameter grid; it does not imply invariance to all parameter changes. IPC omits products across different delays. Of 3,604 chemical edges, 1,864 have unknown or complex signs and are assigned signs to match the labeled-edge ratio; selected controls remove them. Electrical synapses and self-loops are excluded; input reaches all units.

Selected references [1] Fenyves et al. (2020). Synaptic polarity and sign-balance prediction. *PLOS Comput. Biol.* 16, e1007974.
[2] Dambre et al. (2012). Information processing capacity of dynamical systems. *Sci. Rep.* 2, 514.
[3] Roy & Vetterli (2007). The effective rank: a measure of effective dimensionality. *EUSIPCO*, 606–610.